

Selecting between batch and zone order picking strategies in a distribution center

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Abstract

An order picking strategy in a distribution center (DC) defines the manner in which pickers navigate the picking area to pick items from storage locations. We focus on the problem of selecting between a batch picking and a zone picking strategy. For this problem, we propose a cost model to estimate the cost of each type of picking strategy. In our cost model we consider the effects of pick-rate, picker blocking, workload-imbalance, and the sorting system requirement. Through an example problem, we show how system throughput, order sizes, item distribution in orders, and wavelength affect the picking strategy selection decision.

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1. Introduction

Today's supply chains are dependent on effective logistics systems to move products from anywhere in the world to their final destination. And although there is pressure to reduce the inventory in such systems, distribution centers (or alternatively referred to as warehouses, especially when storage is the primary function), where inventory is held until needed at the retail- or customer-level, still play an important role in maintaining high levels of customer service. In the United States alone, a recent Department of Energy study reports that there are nearly 597,000 warehouses in operation, with nearly 15,000 larger than 100,000 ft² and nearly 3000 larger than 500,000 ft² (Energy Information Administration, 2003).

In a distribution center (DC), in addition to storage, the major facility logistics activity is order picking, which refers to the operation of retrieving items from storage locations to fulfill customer orders. Order picking has been identified as the highest priority activity in a DC for productivity improvement due to its relatively high (about 50%) contribution to the total DC operating cost (Tompkins et al., 2003). The

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contribution of this paper is in providing analytical models to aid in the design process of an order picking system.

2. Designing an order picking system

While designing an order picking system, a designer must consider the following question: *which picking system best meets a given set of objectives?* Some of the objectives a designer is required to optimize include maximizing throughput or minimizing either cost, space, response time, or error-rate, or a combination thereof. To answer this question, the designer typically follows the standard engineering design process (Tompkins et al., 2003), which involves the following phases: Phase 1: define the problem; Phase 2: analyze the problem, generate alternatives, evaluate the alternatives, and select the preferred design; and Phase 3: implement the design.

If the designer knows which stock keeping units (a unique identifier of each type of product or item that is stocked in a DC) are to be stored in the picking area and their quantities, then Phase 1 may involve the following problem definitions in an order picking environment:

- What type of storage system should be used?
- Which order picking strategy (i.e., discrete, batch, zone, or bucket brigade) should be used?
- Which picking system (i.e., manual, semi-automated, or automated) should be used?
- Which pick-assist technology (i.e., paper-based or paper-less) should be used?

The first problem has been addressed in Parikh and Meller (2006a). In this paper we focus on the second problem, selecting an appropriate order picking strategy within a medium-to-high throughput DC. To analyze this problem, according to Phase 2 of the engineering design process, the designer first has to identify possible alternatives for this problem. After that, the designer has to devise a proper methodology to select the best performing alternative(s) before moving to Phase 3. The focus of this paper is on evaluating various picking strategies and developing a proper methodology to select the best strategy for a picking system. Our contributions lie in both developing a cost-based evaluation model, as well as defining and solving constituent sub-models for this process.

There are five order picking terms commonly used: discrete, batch, zone, bucket brigade, and wave picking. In *discrete picking* a picker is responsible for picking all the items in a single order during a pick-tour. In *batch picking* several orders are batched (or grouped) together and a picker picks all the items in a given batch. *Zone picking* requires that each picker is assigned to a specific region of the storage area and is responsible for picking the items in that region only. *Bucket brigade picking*, which is actually a control policy for executing discrete order picking, requires that as soon as the most downstream picker completes an order, he/she walks back to take over the order the picker immediately upstream of him/her is currently picking. The latter, in turn, takes over the order of his/her predecessor, and so on until the most upstream picker begins a new order (Bartholdi et al., 2001). Note that, in batch or zone picking, if orders are required to be picked in a predefined time-window (known as a wave), then it is referred to as *wave picking*.

Discrete order picking, though simple to implement, can be labor-intensive for medium-to-high throughput DCs. Moreover, bucket brigade picking is limited to applications where handing-off the items to a downstream picker is easy. It is our observation that most DCs typically do not employ these two strategies; instead they prefer to consider batch or zone picking strategies. Therefore, we consider these two strategies, as illustrated in Fig. 1, in our modeling.

Two types of batch picking strategies exist: pick-and-sort and sort-while-pick (Tompkins et al., 2003). In *pick-and-sort* batch picking the pickers *do not* sort (while picking) the items into customer orders. Such a situation can occur if the cart capacity is too small for the pickers to sort the picked items in the cart. The picked items are, therefore, consolidated downstream through a manual or automated sorting system. This means that pick-and-sort batch picking maintains a high pick-rate (items picked per unit time) for the pickers (as no sorting is involved during picking), but requires a downstream sorter. In contrast, in *sort-while-pick* batch picking the pickers simultaneously pick-and-sort the items into customer orders. Such a situation can occur if the pick-cart capacity is large enough for the pickers to sort the picked items in available compartments in the

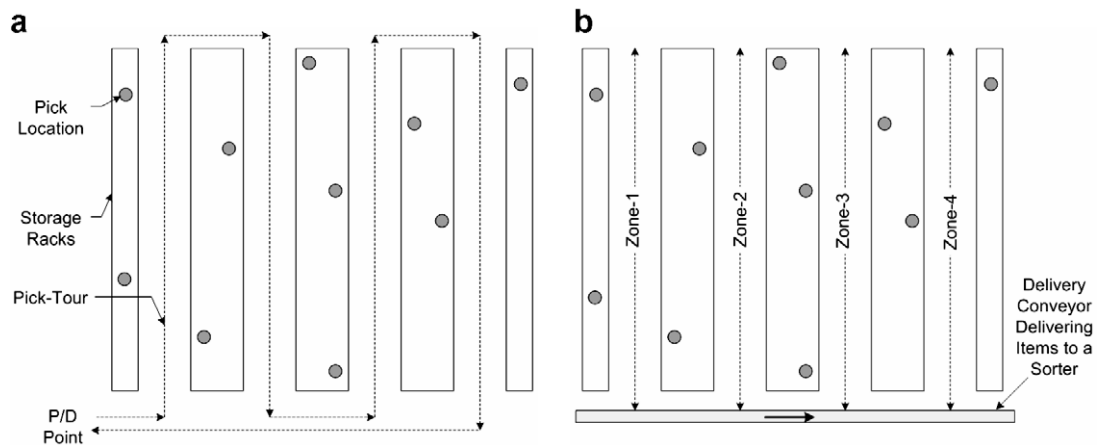


Fig. 1. Schematic of batch and zone picking strategies: (a) batch picking strategy and (b) zone picking strategy.

pick-cart. This means that sort-while-pick batch picking reduces the pick-rate of pickers (due to sorting while picking), but eliminates the need for a downstream sorter. Therefore, there exists a trade-off between pick-rate and the requirement of a sorting system with the two types of batch picking systems.

Note that in both of these types of batch picking systems, pickers travel the entire picking area, which increases their total travel-time. They may also block each other while picking. We elaborate on both of these issues later in this section.

As in batch picking, two types of zone picking strategies exist: sequential and simultaneous (Tompkins et al., 2003). In *sequential* (also known as progressive or pick-and-pass) zone picking, picking occurs one order at a time, and in one zone at a time, after which the order is passed to the next zone. Because only one order is handled at a time, sequential zone picking reduces the pick-rate of pickers, but eliminates the requirement of a downstream sorter. In contrast, in *simultaneous* (also known as synchronized) zone picking, all items corresponding to batched orders are picked simultaneously from all the zones, and then orders are consolidated through a sorting system. Therefore, simultaneous zone picking increases the pick-rate of pickers as the entire batch is picked at a time, but requires a downstream sorter. As a result, with zone picking there also exists a trade-off between pick-rate and the requirement of a sorting system. The sequential and simultaneous zone picking strategies are illustrated in Figs. 2 and 3, respectively.

From the above description, we can see that, at a minimum, *pick-rate* and *the requirement of a sorting system* affect the selection of an appropriate picking strategy within the two classes of picking strategies (batch

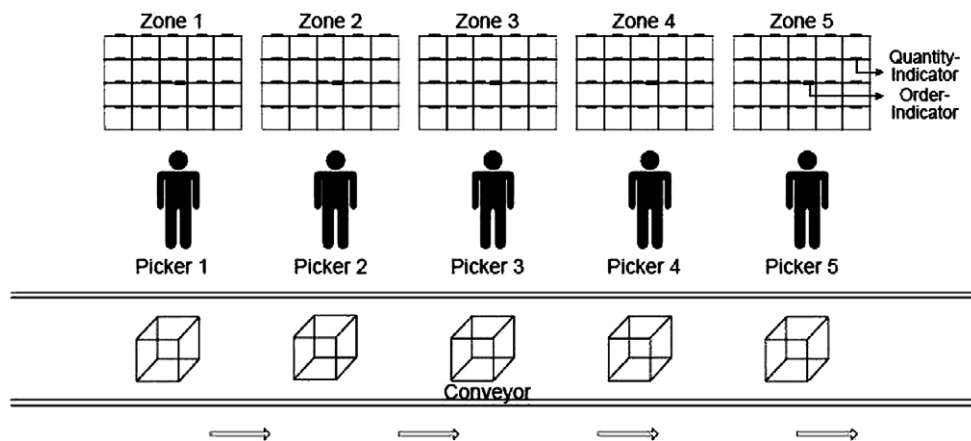


Fig. 2. A sequential zone picking strategy (Jane, 2000).

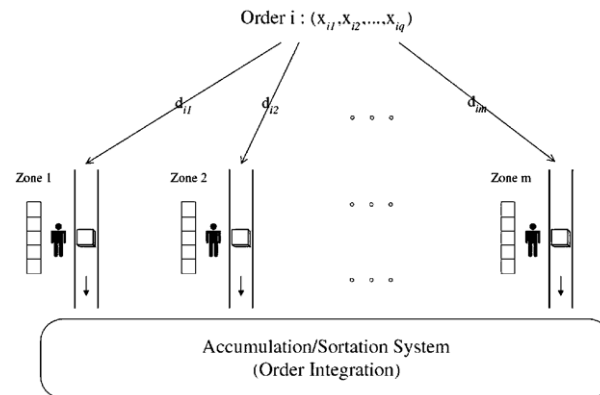


Fig. 3. A simultaneous zone picking strategy (Jane and Lai, 2005).

and zone). In addition, there are two other factors that must be considered when making comparisons between batch and zone systems: *blocking* and *workload-imbalance*. Blocking may be significant in batch picking systems where pickers move freely in traveling to picks. Blocking results in waiting-times, which reduces the productivity of pickers. Additionally, workload-imbalance results when the orders are batched such that unequal workloads are assigned to pickers. Unequal workload may result in orders not being fulfilled in the scheduled hours of operation during the day. If these left over orders are not to affect the next shift's operations, then they will have to be fulfilled through additional labor; e.g., by incurring overtime. Based on our discussions with industry personnel workload-imbalance is an issue more closely related to zone picking systems. Table 1 provides a summary of these factors with respect to the advantages and disadvantages of batch and zone picking types.

In deciding whether to employ a batch or zone picking strategy a designer has to account for each of the four factors just discussed: *pick-rate*, *blocking*, *workload-imbalance*, and *sorting*. We refer to this decision problem as the *batch versus zone problem*. One way to address this problem is to estimate the effect of these four factors and incorporate them into a model to estimate batch or zone picking system cost. The components of the system cost may include *labor*, *equipment*, *imbalance*, and *sorting* costs. Labor cost includes the total annualized cost of labor (i.e., pickers, along with pick-carts, and packers) in the system. Equipment cost includes

Table 1
Summary of advantages and disadvantages of various types of batch and zone picking strategies

Strategy	Type	Advantages	Disadvantages
Batch picking	Pick-and-sort	- Increases pick-rate of pickers as sorting not part of picking - Reduces chances of workload-imbalance	- Requires a sorting system - Increases probability of blocking - Decreases pick-rate of pickers as pick-tours are long
	Sort-while-pick	- Reduces chances of workload-imbalance - Does not require a sorting system	- Decreases pick-rate as sorting part of picking process - Decreases pick-rate of pickers as pick-tours are long - Increases probability of blocking
Zone picking	Progressive	- Increases pick-rate of pickers as pick-tours are short - Does not require a sorting system - Eliminates blocking	Increases chances of workload-imbalance
	Synchronized	- Increases pick-rate of pickers as pick-tours are short - Eliminates blocking	- Requires a sorting system - Increases chances of workload-imbalance

cost of racks, conveyors, etc. Imbalance cost is the cost incurred to fulfill items that were not fulfilled during the scheduled hours of operation in a day; this cost component is explained in more detail later. Sorting cost is the annualized cost of a sorting system, a model for which we develop later.

To illustrate how the four factors mentioned earlier affect the four cost components, assume that a designer is required to select between a sort-while-pick batch picking system and a simultaneous zone picking system. Because sorting is part of the picking process in sort-while-pick batch picking, the pick-rate of the pickers is relatively low. Therefore, the labor cost in sort-while-pick batch picking is high relative to simultaneous zone picking. The equipment cost in simultaneous zone picking may be high if take-away conveyors and additional technology (e.g., pick-to-light) is employed. Excessive workload-imbalance in simultaneous zone picking, compared to sort-while-pick batch picking, will increase the number of orders not fulfilled during the day, which increases the imbalance cost. Furthermore, simultaneous zone picking requires a sorter to sort items into orders, which results in the cost of a sorter. (Sort-while-pick batch picking does not require a sorter.) Therefore, the designer will have to estimate all four cost components (labor, equipment, imbalance, and sorting) before deciding which one of the two picking strategies, and which version of it, should be employed. Our aim in this paper is to address the batch versus zone problem by developing a cost model that is appropriate for use in evaluating either type of picking strategy.

The remainder of this paper is organized as follows. In Section 3 we briefly review literature in the area of picking system design. In Section 4 we present a cost model to estimate the cost of batch picking and zone picking systems. As mentioned earlier, since our cost model requires estimating costs due to workload-imbalance and a sorting system, in Sections 5–8, respectively, we present methods for doing so. In Section 9 we revisit our cost model for the batch versus zone problem through an illustrative example. Finally, in Section 10 we summarize our understanding of the batch versus zone problem.

3. Literature review

In the area of overall DC (or warehouse) design, Gray et al. (1992) propose a multi-stage hierarchical decision approach to model the composite design and operating problems for a typical order-consolidation warehouse. Their hierarchical approach utilizes a sequence of coordinated mathematical models to evaluate the major economic trade-offs and to prune the decision space to a few superior alternatives. Rouwenhorst et al. (2000) present a reference framework and a classification of warehouse design and control problems. They define a warehouse design problem as a “structured approach of decision making at a strategic, tactical, and operational level in an attempt to meet a number of well-defined performance criteria.”

In the area of design of a picking system in a DC, Brynzer et al. (1994) present a zero-based analysis methodology for the evaluation of OPSs as a base for system design and managerial decisions. Yoon and Sharp (1996) propose a structured procedure for the analysis and design of a picking system that considers the inter-dependent relationships between different functional areas (e.g., receiving, picking, sorting, etc.). The survey papers by van den Berg (1999), Rouwenhorst et al. (2000), and Gu et al. (2005) provide a complete review of various contributions in the area of DC and picking system design.

The problem of selecting an order picking strategy is critical as it affects the design of aisles and storage systems, picker-routing, etc., all of which affect the throughput of the system. The primary objective of a picking strategy is to maximize throughput, or minimize cost (or response time).

Petersen (2000) states that the choice of a picking strategy can have a tremendous effect on the efficiency and cost of a picking system in mail order companies. To this end, he evaluates five order picking strategies—discrete (or strict), batch, sequential (or pick-and-pass) zone, simultaneous zone (which he calls batch-zone), and simultaneous zone-wave—using a simulation model. Based on the results, Petersen concludes that simultaneous zone-wave picking and batch picking are superior, and that their performance is not adversely affected by changes in demand skewness patterns or daily order volume. On the other hand, the author notes that the performance of sequential zone and batch-zone picking deteriorates as order volume increases.

Characterizing bucket brigade picking as a self-balancing strategy (and a control policy for discrete order picking), Bartholdi et al. (2001) develop stochastic models to analyze the performance of this strategy in high-volume OPSs. Through simulation experiments comparing bucket brigade picking to sequential zone picking,

they suggest that the production rate efficiency (a ratio of realized production rate to maximum possible rate) of a simulated bucket brigade is similar to that of zone picking for the case when the picking system had identical pickers (in terms of their walk and pick-times). As the difference in the walk and pick-times of individual pickers increases, bucket brigade picking is more productive. The limitation of this strategy is that it is most effective when the picking aisle is structured as a flow line, and when the time to hand-off an order is low.

Jane (2000) considers a sequential zone picking system, which he refers to as a *relay* picking system. He addresses the problem of assigning n products into m storage zones (one picker per zone) with the objective of minimizing the differences that might exist between each picker's total number of picks. Jane and Lai (2005) propose a clustering algorithm for item assignment in a simultaneous zone picking system. The authors propose a similarity measure between any two items for measuring the co-appearance of both items in the same order. Accordingly, items frequently ordered together are located in different zones to minimize the idle time in the simultaneous zone system.

Recently, there has been some work related to the batch versus zone problem. For example, Russell and Meller (2003) develop cost and throughput models for manual and automated sorting systems. Sorting systems are required in pick-and-sort batch picking and simultaneous zone picking systems, and, as described earlier, play a major role in the batch versus zone problem. Their research is aimed at providing a design aid to decide whether or not to automate the sorting process. They develop a descriptive model for the sorting system design decision based on demand levels, labor rates, order sizes, and other factors, and incorporate the descriptive model into a cost-based optimization model to recommend a solution.

Gue et al. (2006) address the effect of pick density on order picking areas in narrow-aisle DCs. Since a batch picking strategy may induce picker blocking, which can reduce system throughput, they develop analytical models to estimate *in-the-aisle* picker blocking in a narrow-aisle DC. Using discrete-time Markov chains, the authors estimate bounds on the percentage of time each picker is blocked for the case when pickers pick only one item from a pick-column when stopped. Parikh and Meller (2006b) suggest that in many DCs, the aisles are wide enough to permit passing. Although blocking is less in such DCs, the pickers may still experience blocking whenever two or more pickers need to pick at a pick-column at the same time. They refer this form of blocking as *pick-column* blocking. Similar to Gue et al. (2006), they employ a discrete-time Markov chain approach to estimate the average picker blocking in wide-aisle DCs. Parikh and Meller (2006b) also consider the case when there is a high variance in the time spent picking once stopped (which is likely to occur when multiple picks may be possible at a single stop). As mentioned earlier, picker blocking can play a significant role in the batch versus zone problem.

To our knowledge there has been no research that addresses the batch versus zone problem. We next present a possible approach to address this problem with a cost model-based approach.

4. A cost model for the batch versus zone problem

We model the batch versus zone problem on the basis of cost. Accordingly, we develop a cost model to estimate the cost of picking systems employing batch and zone picking strategies. The same model may be used to select between alternate versions of either strategy, a point we revisit in Section 9.

The cost model includes the expected costs of pickers, equipment, imbalance, sorting, and packers. (Note that we represent the expected cost of labor as the expected cost of pickers and packers and, for simplicity, we use x instead of $E[x]$ throughout when representing the expected value of x .) Let C^b and C^z represent batch and zone picking system costs, respectively, which can be estimated as follows:

$$C^b = P^b c_p + c_e^b + U^b c_u + c_s(\alpha^b I) + K^b c_k \quad \text{and} \quad C^z = P^z c_p + c_e^z + U^z c_u + c_s(\alpha^z I) + K^z c_k, \quad (1)$$

where

- I = expected number of items to be picked,
- P^b (P^z) = expected number of pickers required in batch (zone) picking,
- c_p = annualized loaded cost of a picker (\$); including the cost of a picking vehicle,
- c_e^b (c_e^z) = annualized equipment cost in batch (zone) picking (\$); including the cost of racks, conveyors, etc.,

- U^b (U^z) = expected number of items in orders unfulfilled due to workload-imbalance in a day in batch (zone) picking,
- c_u = yearly unit imbalance cost to fulfill an item (\$/unit/year); including the costs associated with overtime of pickers, packers, sorter, shipping personnel, etc.,
- K^b (K^z) = expected number of packers required in batch (zone) picking to pack the picked (and sorted) items into shipping cartons before sending these cartons to shipping,
- c_k = annualized loaded cost of a packer (\$),
- α^b (α^z) = binary parameter for batch (zone) picking; 1, if sorter required, and 0, otherwise; $\alpha^b = 1$ for pick-and-sort batch picking, while $\alpha^z = 1$ for simultaneous zone picking, and
- $c_s(I)$ = annualized cost of a sorting system to sort I ($I > 0$) items (\$); includes the cost of a sorting conveyor, sort-lanes (with diverting mechanisms), induction stations, and inductors.

Some of the cost components in (1) are relatively straightforward to estimate, while others are more difficult to estimate. For example, estimating the costs associated with pickers, equipment, and packers is not difficult. However, estimating the imbalance and sorting system costs is difficult. Therefore, in the next two sections we develop models to estimate these two cost components. Once all the cost components are estimated, the designer can use (1) to estimate the cost of employing either a batch picking or a zone picking strategy. We define the best picking strategy as the one that corresponds to the minimum of the two system costs, C^b and C^z ; i.e., if C represents the cost of the least expensive picking system, then the batch versus zone problem can be defined logically as

$$\begin{aligned} & \text{minimize} && C \\ & \text{subject to} && C = \min\{C^b, C^z\}. \end{aligned}$$

5. Estimating the imbalance cost in picking systems

In picking systems it is important that the pickers are assigned a workload that is nearly the same for all pickers within and across all waves. A balanced workload will ensure that all pickers finish their picking at the end of the wave. The advantage of doing so, in addition to minimizing the labor cost, is the efficient scheduling of sorting and shipping operations. Moreover, a balanced workload also helps to ensure that all customer orders are fulfilled in the scheduled hours of operation during the day so that there is no overtime, and hence, eliminating the imbalance cost.

We use the total number of items picked as a surrogate for workload (as opposed to the total time for a picker directly). Our reasoning for this is as follows: the total time for a picker is the sum of total travel-time, which depends on the routing of pickers, and total pick-time. Since we are mostly concerned with medium-to-high throughput DCs, we assume here that the pickers pass by each pick location once during a pick-tour (Gue et al., 2006); therefore, the total travel-time is fixed, and we can eliminate it from our model. This leaves us with pick-time, which depends on the number of items picked. Furthermore, similar to Gue et al. (2006), we assume that pick-time is linear in the number of items picked.

Workload-imbalance occurs when the workload assigned amongst pickers is not equal. To illustrate this point further, consider an example of a picking area that consists of 2 aisles in which 2 pickers perform the picking operation. The system operates two waves, and each picker can pick a maximum of 10 items in a wave. Assume that 10 orders (with a total of 40 items) have to be fulfilled in a day. The distribution of items within these 10 orders over 2 aisles is illustrated in Table 2.

For this example, if a batch picking system is employed (where both pickers travel the 2 aisles), then we need to only focus on the last column in Table 2. A possible way in which the 10 orders can be batched to reduce workload-imbalance between the 2 pickers is illustrated in Table 3. In the table we observe that all 10 orders have been assigned among the 2 pickers, and both the pickers have been assigned a balanced workload of 10 items in each wave. That is, all the orders have been assigned, and the workload amongst the pickers is balanced.

Now consider a zone picking system, where each aisle is a zone and one picker is assigned to each zone. We therefore now need to focus on the middle two columns in Table 2. A possible way by which orders can be

Table 2
Distribution of items in the 10-order example

Order #	Aisle 1	Aisle 2	Total items
1	1	2	3
2	5	0	5
3	2	3	5
4	2	0	2
5	0	4	4
6	1	4	5
7	3	2	5
8	0	4	4
9	2	2	4
10	1	2	3
Total	17	23	40

Table 3
A possible order batching solution for the batch picking system for the 10-order example

		Orders assigned	Total items
Wave 1	Picker 1	2, 3	10
	Picker 2	6, 7	10
Wave 2	Picker 1	1, 9, 10	10
	Picker 2	4, 5, 8	10
Total			40

batched to minimize workload-imbalance between the 2 pickers is illustrated in Table 4. Notice that a total of 9 out of 10 orders have been assigned during the two waves. Order #8 is not assigned because it could not be accommodated during either of the waves. That is, if this order were included in one of the waves, then it would violate the capacity of the pickers in Zone 2 (which is 10 items). As a result, Order #8 will have to be picked during overtime. That is, not all orders have been assigned, and the workload amongst zones is not balanced.

In general the following two factors affect the process of generating a balanced workload for pickers: (i) item to zone assignment; and (ii) order batching. Considering (i), if the items are inappropriately assigned to zones, then it implies that a few zones are assigned a larger pool of frequently requested items than other zones. As a result, when the orders are batched, the pickers in these few zones may be required to pick a larger number of items than the pickers in other zones. This results in workload-imbalance. Considering (ii), if the orders are inappropriately batched, then more workload may be assigned to some pickers than others. This may be because of an inappropriate policy used for batching orders. This results in workload-imbalance. Workload-imbalance, due to either (i) and/or (ii), may result in customer orders not being fulfilled during the scheduled hours of operation in a day. We, therefore, use the number of items (in customer orders) unful-

Table 4
A possible order batching solution for a zone picking system

		Orders assigned	Total items
Wave 1	Zone 1	1, 2, 5, 9, 10	9
	Zone 2	1, 2, 5, 9, 10	10
Wave 2	Zone 1	3, 4, 6, 7	8
	Zone 2	3, 4, 6, 7	9
Total			36

filled as a measure of workload-imbalance and refer to the cost of fulfilling these unfulfilled items as the *imbalance cost*.

In this paper we assume that the items are appropriately assigned to zones using the approach presented in Jane and Lai (2005). Therefore, we eliminate reason (i) listed above, and focus on (ii), which relates to the order batching policy used. We also assume that the order pool is static: i.e., the order pool contains only those orders received before a certain time in the day.

To quantify the effects of workload-imbalance for batch and zone picking strategies on equal footing, we formulate mathematical models to batch orders. With these models, we can identify the orders that were not fulfilled during the scheduled hours of operation (which determines the imbalance cost). We use a dual bin-packing problem analogy to discuss our model formulations.

6. Order batching models

The classical bin-packing problem (BPP) is defined as follows: *given a number of items of integer sizes, what is the minimum number of bins, each having the same integer capacity, necessary to assign all items?* We can draw an analogy between our problem and the BPP by considering that (i) a bin represents a picker, (ii) the capacity of the bin represents the predefined capacity of the picker in terms of the number of items that can be picked, and (iii) the assignment of items to bins represents the assignment of orders to pickers. Therefore, we can define the order batching problem (based on the BPP) as follows: *given a number of orders consisting of items (integers), what is the minimum number of pickers, each having the same integer capacity, necessary to fulfill all orders?* This analogy suggests that the number of pickers is a decision variable. That is, if the number of orders to be fulfilled changes daily, then the number of pickers may also change daily. As the hiring and firing of pickers on daily basis is generally not feasible, most DCs fix the number of pickers for a certain time-period based on the expected demand for that time-period. Hence, the BPP analogy does not hold completely for the order batching problem.

Instead, formulating the order batching problem analogous to the dual bin-packing problem (dual BPP) seems appropriate. The dual BPP is defined as follows: *given a fixed number of capacitated bins, what is the maximum number of items that can be packed?* The analogy between items and orders, and bins and pickers still holds; however, the definition of the order batching problem (based on the dual BPP) changes and is as follows: *given a fixed number of pickers, what is the maximum number of items that can be fulfilled?* In the order batching problem considered here we are given a fixed number of pickers picking in a given number of waves. Each picker can pick no more than a certain fixed number of items during each wave. The objective of the order batching problem is to balance the workload amongst pickers such that the number of items fulfilled during the scheduled hours of operation is maximized. That is, the order batching problem we consider follows very closely to the dual BPP. We assume that any unfulfilled items (in orders) during the scheduled hours of operation will need to be fulfilled by incurring overtime, which will increase the imbalance cost.

First consider the order batching problem for batch picking, which we will use to identify the optimal assignment of orders to pickers in a batch picking system. To facilitate model formulation we use the number of items as a surrogate for the workload of each picker. Let

- i = index for orders; $i = 1, \dots, D$,
- j = index for pickers; $j = 1, \dots, P$,
- k = index for waves; $k = 1, \dots, W$,
- d_i = parameter for the number of items in order i ,
- M^b = parameter for the maximum number of items that can be picked by a picker in a wave in a batch picking system, and
- $x_{ijk} = \begin{cases} 1, & \text{if order } i \text{ is assigned to picker } j \text{ in wave } k, \\ 0, & \text{otherwise.} \end{cases}$

Note that the parameter M^b depends on the expected pick-rate of the pickers; i.e., the number of items picked in unit time. The expected value of pick-rate depends on the following three factors: (i) pick-time, (ii) travel-time, and (iii) pick-cart capacity. The expected value of pick-time can be obtained through time-studies. The

expected value of travel-time depends on the picker-routing policy (e.g., traversal, largest gap, return, etc.) and can be estimated through models proposed in the order picking literature (e.g., Hall, 1993; Ratliff and Rosen-thal, 1983; or Roodbergen and Vis, 2006). The pick-cart capacity is assumed to be a fixed constant depending on the type of the pick-cart used and the sizes of items picked. For simplicity, we assume that the expected value of pick-rate of the pickers has already been established through time-studies after accounting for factors (i), (ii), and (iii).

The mathematical model for the order batching problem for batch picking (labeled OBP-BP) is as follows:

$$\text{OBP-BP: maximize} \quad \sum_{i=1}^D \sum_{j=1}^P \sum_{k=1}^W d_i x_{ijk} \quad (2)$$

$$\text{subject to} \quad \sum_{i=1}^D d_i x_{ijk} = b_{jk} \quad \forall j, k, \quad (3)$$

$$b_{jk} \leq M^b \quad \forall j, k, \quad (4)$$

$$\sum_{j=1}^P \sum_{k=1}^W x_{ijk} \leq 1 \quad \forall i, \quad (5)$$

$$x_{ijk} \in \{0, 1\} \quad \forall i, j, k.$$

In the above formulation the decision variable is x_{ijk} ; i.e., the assignment of order i to picker j in wave k . We calculate the current number of items assigned to each picker in a wave (i.e., b_{jk}) through (3). In (4) we ensure that no b_{jk} value exceeds the capacity of a picker (M^b). We use (5) to ensure that no order is assigned to more than one picker in one of the W waves. That is, an order i , if assigned to a wave k , may be assigned to no more than one picker j . The objective, represented by (2), is to maximize the number of items fulfilled during the scheduled hours of operation.

By solving the model OBP-BP we can obtain the values of b_{jk} , which provides the current workload of each picker during a wave. The orders with all of their x_{ijk} values equal to zero could not be included in the batches formed, and hence will not be fulfilled during the scheduled hours of operation. For use in our cost model objective function, (1), the total number of items corresponding to these unfulfilled orders (U^b) can be calculated as the difference between the total items to be picked and the OBP-BP objective function value. That is,

$$U^b = \sum_{i=1}^D d_i - \sum_{i=1}^D \sum_{j=1}^P \sum_{k=1}^W d_i x_{ijk}.$$

Next we consider the order batching problem for zone picking in which we assume that only one picker is assigned to each zone, and modify the OBP-BP model, to identify the optimal assignment of orders to pickers in a zone picking system. There are two key differences between the batch and zone picking models: (i) along with the number of items in an order, we need to consider the distribution of items over the zones in that order, and (ii) we only need to consider the assignment of orders to waves, because when an order is assigned to a wave, it is automatically assigned to all the applicable zones from where the items within that order are to be picked. Accordingly, let

- j = index for zones; $j = 1, \dots, Z$,
- d_{ij} = parameter for the number of items in order i to be picked from zone j ,
- M^z = parameter for the maximum number of items that can be picked by a picker in a wave in a batch picking system, and
- $x_{ik} = \begin{cases} 1, & \text{if order } i \text{ is assigned to wave } k, \\ 0, & \text{otherwise.} \end{cases}$

Note that the values of M^b and M^z may be different as the pick-rates of pickers in a batch picking and a zone picking system are likely to be different (as illustrated later).

A mathematical model for the order batching problem for zone picking (labeled OBP-ZP) is as follows:

$$\text{OBP-ZP: maximize } \sum_{i=1}^D \sum_{j=1}^Z \sum_{k=1}^W d_{ij} x_{ik} \quad (6)$$

$$\text{subject to } \sum_{i=1}^D d_{ij} x_{ik} = b_{jk} \quad \forall j, k, \quad (7)$$

$$b_{jk} \leq M^z \quad \forall j, k, \quad (8)$$

$$\sum_{k=1}^W x_{ik} \leq 1 \quad \forall i, \quad (9)$$

$$x_{ik} \in \{0, 1\} \quad \forall i, k.$$

In this formulation the decision variable is x_{ik} , i.e., the assignment of order i to wave k . The current number of items assigned to each zone in a wave (b_{jk}) is specified using (7). In (8) we ensure that no b_{jk} value exceeds the capacity of a zone (M^z). We use (9) to ensure that no order is assigned to more than one wave. The objective, represented by (6), is to maximize the total number of items fulfilled.

Similar to the OBP-BP model, by solving the OBP-ZP model we can obtain the values of b_{jk} , which provides the current workload of each picker during a wave. For use in our cost model objective function, (1), the total number of items corresponding to the unfulfilled orders (U^z) can be calculated as the difference between the total items to be picked and the objective function value. That is,

$$U^z = \sum_{i=1}^D \sum_{j=1}^Z d_{ij} - \sum_{i=1}^D \sum_{j=1}^Z \sum_{k=1}^W d_{ij} x_{ik}.$$

Note that the number of binary decision variables (x_{ik}) in the OBP-ZP model is smaller than the number of binary decision variables (x_{ijk}) in the OBP-BP model. This is because in zone picking an order is assigned directly to wave k , while in batch picking an order is assigned to a specific picker j in a wave k . Consequently, the OBP-BP model may be considerably harder to solve as compared to the OBP-ZP model, as we illustrate in Section 9.

We next, through a small experimental study, illustrate the use of our order batching models to estimate workload-imbalance (in terms of the number of items unfulfilled).

7. Observations from an experimental study

In Appendix A we present an experimental study, the purpose of conducting was to (1) determine the relative performance of the OBP models as compared to simple non-optimization-based policies for batch picking and zone picking systems; (2) explore the runtimes required for the OBP models; (3) confirm that workload-imbalance is more of an issue in zone picking systems; (4) measure the impact of item distribution across the zones for a zone picking system; and (5) examine the impact of zone length on the solutions to the OBP models.

In addition to obtaining the optimal solution to the order batching problem using the OBP-BP and OBP-ZP models, we also included the following four heuristic order batching policies:

1. a first-come-first-serve (FCFS) order batching policy, in which orders are batched in the order they are received, with the batching initiated by first considering the lowest indexed picker (or wave);
2. a 50-order look-ahead FCFS policy (FCFS-50), in which the FCFS order batching policy is used and whenever an order cannot be allocated to a picker (or wave) because of the picker-capacity (M^b or M^z) constraint, the subsequent 50 orders are checked for inclusion in the current batch;
3. a 200-order look-ahead FCFS policy (FCFS-200), in which, instead of checking the subsequent 50 orders, the subsequent 200 orders are checked for inclusion in the current batch; and
4. a rule similar to first-fit-decreasing (FFD) policy typically employed for solving BPPs and dual BPPs, in which the orders are first sorted in decreasing order in terms of number of the items, and then each successive order is allocated to a picker (or wave) starting from the lowest indexed picker (or wave).

The following observations represent what was learned from our experimental study. Full details are provided in [Appendices B and C](#).

Observation 1: In terms of the number of items unfulfilled in the order batching problem for batch and zone picking, the FFD policy performs at least as well as, and in most cases out-performs, the other three policies, FCFS, FCFS-50, and FCFS-200, for the two order sizes and the two item distributions. Moreover, among the various FCFS policies the FCFS policy performs the worst, while the FCFS-200 policy never performed worse than the FCFS-50 policy, a result that we formalize below.

Theorem 1. *The number of items fulfilled (i.e., the objective function value in the OBP-BP and OBP-ZP models, as presented in (2) and (6)) using the FCFS-200 policy is never less than the number of items fulfilled using the FCFS-50 policy.*

Proof. By contradiction. Accordingly, we assume that the number of items fulfilled (i.e., the objective function value in the OBP-BP and OBP-ZP models, as presented in (2) and (6)) using the FCFS-50 and FCFS-200 policies follow the relationship

$$N_{200} < N_{50}, \quad (10)$$

where N_{50} and N_{200} are the objective function values obtained using the FCFS-50 and FCFS-200 policies, respectively.

Note that both the FCFS-50 and FCFS-200 batching policies start assigning orders to the current batch in the same way as the FCFS policy. Let N represent the objective function value of the current batch when using the FCFS-50 and FCFS-200 policies just before identifying that the next order, say d_v , cannot be assigned to the current batch in the current wave as it will violate the batch-capacity.

When the order d_v is encountered, the subsequent 50 and 200 orders for FCFS-50 and FCFS-200, respectively, are scanned. Let n_{50} and n_{200} represent the number of items in orders found in the subsequent 50 and 200 orders, respectively, starting from order number d_v , that fit in the current batch. Then, $N_{50} = N + n_{50}$ and $N_{200} = N + n_{200}$. Therefore, from (10),

$$n_{200} < n_{50}. \quad (11)$$

As the next 50 or 200 orders (depending on the policy used) are scanned only once before the current batch is considered *completely assigned* (after which it is removed from further consideration for the batch), the set of orders $\{1, 2, \dots, 50\}$ is actually a *subset* of the set of orders $\{1, 2, \dots, 50, \dots, 200\}$. Therefore, $n_{200} \geq n_{50}$, which contradicts (11).

This basic argument extends to the subsequent waves as well. $\square$

Observation 2: In terms of solution time, all four policies (FCFS, FCFS-50, FCFS-200, and FFD) take less than 20 s on a standard personal computer to solve the order batching problems during the study. While all instances of the OBP-ZP model for small-sized orders were solved optimally in a reasonable time (less than 305 s), a few instances for large-sized problems could not be solved optimally even after 24-h of processing. In contrast, except for one instance no other OBP-BP model for small-sized or large-sized orders could be solved optimally after 24-h of processing. As was highlighted earlier, this is likely due to the number of binary variables in the OBP-BP model, which is larger than in the OBP-ZP model. Although a large number of binary decision variables in the OBP-BP model provides flexibility in assigning orders to pickers, and thus, decreases the number of unfulfilled items, solving such a problem is difficult. In contrast, the number of binary variables in the OBP-ZP model is relatively small, which decreases the flexibility of the model to assign orders to pickers. As a result, the number of items unfulfilled in a zone picking system is relatively high.

The FFD policy consistently achieves the optimal (or near-optimal) solution to the order batching problem for both the batch and zone picking systems when the order sizes are small. However, as the order size increases the solution gap between the FFD policy and the optimal (or best) solution obtained by solving the OBP-BP and OBP-ZP models increases to a point where the gap may be a motivation to use the OBP optimization models. (Nevertheless, due to the runtime issues presented above, the remaining observations are therefore based on the solutions obtained from the FFD policy and/or the OBP optimization models, as available.)

Observation 3: The number of items unfulfilled due to workload-imbalance is higher in a zone picking system than in a batch picking system for various order sizes, item distributions, and number of waves. For the

small-sized order problems we solved, there were never any items remaining unfulfilled in the batch picking system; however, the number of unfulfilled items were high in the zone picking system depending on the distribution of items in orders. For large-sized orders, workload-imbalance affects both the batch picking and zone picking systems; however, this effect is always greater in the latter.

Observation 4: The distribution of items included in orders over the picking area has a major effect on the number of items unfulfilled in zone picking systems. In our study we observed that the number of items unfulfilled in orders generated using a skewed distribution is higher than using a uniform distribution. As noted earlier, item distribution in the orders does not impact the order batching problem for batch picking systems.

Observation 5: For small-sized orders, the number of items unfulfilled is unaffected by changes in the number of waves. However, for large-sized orders, as the number of waves increases, the number of items unfulfilled increases (nearly linearly).

Though we observed that the FFD policy performed very well when assigning items to pickers, the solution obtained does not guarantee that all pickers will be assigned a nearly equal number of orders. That is, in an operational sense, a few pickers will be assigned a relatively small number of orders that each contain a large number of items, while the other pickers will be assigned a relatively large number of orders that each contain a small number of items. It is possible to devise a greedy local-improvement algorithm that would begin with the order-to-picker assignment obtained through the FFD policy and then perform a local search to balance the assignment of orders to pickers without changing the original objective function value (i.e., the total number of items assigned to pickers). In so doing, our objective will be to obtain a batching that has a nearly equal number of items, and a nearly equal number of orders, assigned to each picker.

As stated at the outset, the workload-imbalance cost issue is only one issue that must be considered in the batch versus zone problem. Now that we have this issue of the problem modeled, the other issue needing further study is the cost of a sorting system.

8. Estimating the sorting system cost

Our next objective is to model the cost of an automated sorting system, which is comprised of a sorting conveyor, sort-lanes (with diverting mechanisms), one or more inductors at induction stations, and hardware and software controls. A sorting conveyor is required to transfer the items to the sort-lanes. The sort-lanes are indexed according to customer order(s) and typically employ gravity chutes or powered roller conveyors to carry the sorted item(s) to packers. The packers pack these orders into shipping cartons and then send them to shipping. A diverting mechanism at the sort-lane helps to divert items on the sorting conveyor into the sort-lane. This can be achieved either through a pivot arm, pop-up wheels, a tilt-tray, a bomb-bay tray, a cross-belt conveyor, or sliding shoes. An induction station consists of one or more inductors (manual or automatic) that place items onto the sorting conveyor. The hardware and software controls associate the incoming item with the sort-lane, execute the sorting operation, and ensure proper system operation.

We estimate the cost of a sorting system as the sum of fixed and variable costs. The fixed cost of a sorter is the cost of the sorting conveyor and associated hardware and software controls. The variable cost depends on the number of induction stations, inductors per induction station, and sort-lanes (with diverting mechanisms). Therefore, the cost of a sorting system ($c_s(I)$) can be estimated as follows:

$$c_s(I) = c_f + n_l c_l + n_s c_s + n_i c_i, \quad (12)$$

where

- c_f = fixed cost of the sorter (\$),
- n_l = number of sort-lanes required,
- c_l = annualized cost of a sort-lane (\$),
- n_s = number of induction stations required,
- c_s = annualized cost of an induction station (\$),
- n_i = number of inductors required, and
- c_i = annualized loaded cost of labor (\$).

Note that our model is similar to the one presented in [Russell and Meller \(2003\)](#). However, our contribution is two-fold:

1. In our model we consider the cost of sort-lanes. This is important since the number of sort-lanes impacts the system cost and to not consider the sort-lanes would lead to an incomplete model.
2. We develop our model from the point-of-view of the number of orders that needs to be sorted and not by the components in the sorter. That is, by evaluating the sorter cost over the entire range of orders that a sorter (with a given maximum conveyor speed) is capable of sorting, we can accurately reflect the economies-of-scale and diminishing returns inherent to a sorter. By focusing on what the sorter does (sort orders) instead of how it does this (with components), we ensure that the costs can be accurately represented to specify sorter cost a priori of making a decision on the size of the sorter to be used. This aspect is explained through a parametric cost function later in this section.

Note that in a sorter the number of sort-lanes depends on the total number of orders to be sorted and the number of orders assigned to each sort-lane. That is, if D^w represents the total number of orders to be sorted in a wave and d_l represents the number of orders that can be assigned to a sort-lane, then the number of sort-lanes required (n_l) can be estimated as, $n_l = \left\lceil \frac{D^w}{d_l} \right\rceil$.

To estimate the number of induction stations (n_s) and inductors (n_i) required for the sorting system, we adopt the approach presented in [Russell and Meller \(2003\)](#). We summarize key expressions from their work here. Let y represent the total induction rate in items/min; $y = (D^w I_{\text{avg}})/60$, where D^w and I_{avg} represent the number of orders in a wave and the average number of items in an order, respectively. Let s represent the speed of the sorting conveyor (in items/min) and λ represent the nominal induction rate of the inductors (in items/min). Therefore, the total number of induction stations (n_s) can be estimated as

$$n_s = \begin{cases} 1, & \text{if } y \leq s \\ \left\lceil \frac{y}{s-y} \right\rceil, & \text{otherwise.} \end{cases}$$

The total number of inductors (n_i) can be estimated using the model developed by [Johnson and Meller \(2002\)](#). Using their geometric model, [Russell and Meller \(2003\)](#) constructed a table (Table A1 in the Appendix of their article) to estimate the induction rate of each inductor at an induction station. They assume that at most four identical induction stations may be used with no more than four inductors at each induction station. Using the appropriate row in their table, the effective rate of each inductor as a percentage of the conveyor speed can be determined. This percentage, when multiplied by the conveyor speed, determines the actual induction rate (in items/min) of each inductor at each induction station. The number of inductors at each station can then be estimated by adding the number of items for the first inductor, then the second inductor, and so on, until the required number of items to be inducted at the station is reached (i.e., y/n_s). The total number of inductors (n_i) is the sum of inductors at all n_s stations.

For example, let $y = 50$ items/min, $s = 100$ items/min, and $\lambda = 20$ items/min. As $y < s$, $n_s = 1$; i.e., the number of induction stations required is equal to 1. With a nominal induction rate of 20 items/min per inductor, the nominal rate as a percentage of conveyor speed is 20% (20/100). From the Table A1 presented in [Russell and Meller \(2003\)](#), we observe that the induction rates (in %) for the four possible inductors at the induction station are 20.00%, 19.05%, 17.73%, and 15.84%, respectively; the corresponding induction rates (in items/min) are 20.00, 19.05, 17.73, and 15.84, respectively. Consequently, to satisfy the required total induction rate ($y = 50$ items/min), three inductors are required ($20.00 + 19.05 + 17.73 = 56.78 > 50$).

Thus, we can estimate the required sorting system parameter values, and consequently, the cost of the sorting system using (12). Fig. 4 illustrates a parametric cost function to estimate the cost of a sorting system, where l represents the minimum increments of sort-lanes to consider; an actual cost function is illustrated in Section 9.

In Fig. 4 the term $c_f + lc_l + c_s + c_i$ indicates the initial cost of the sorter installed with l sort-lanes and one induction station with one inductor. Depending on the nominal induction rate of inductors and the conveyor speed such a configuration may be suitable for sorting a certain number of orders (say, d_1). If more orders were to be sorted (say, d_2), then additional sort-lanes will be required, the cost of which may be estimated as lc_l .

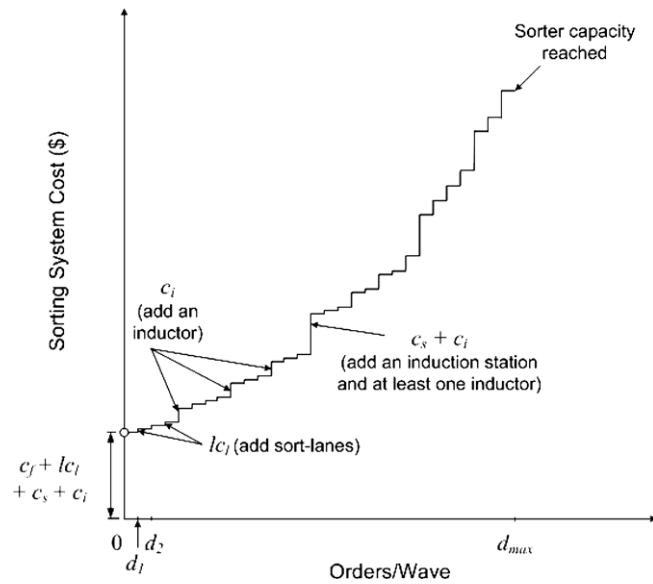


Fig. 4. A parametric cost function to estimate the cost of a sorting system.

If the orders to be sorted increases further, it is possible that one inductor at the induction station may not be able to meet the total induction rate of the system (y). This means that more inductors will be required at that induction station; this is indicated by the cost component c_i in Fig. 4. If the sum of the induction rates of all inductors at an induction station is greater than the conveyor speed (s) but less than the total induction rate (y), then a second induction station will need to be employed; this is indicated by the term $c_s + c_i$ in Fig. 4. Note that at least one additional inductor will be required and that reallocation of the inductors across all induction stations will occur. The process of adding sort-lanes, inductors, and induction stations will continue until the total induction rate (y), corresponding to orders to be sorted in a wave, is met.

Note that as with any other system, the law of diminishing returns applies to the sorting system as well. Accordingly, the additional number of orders that can be sorted by adding the second induction station is less than the number of orders that can be sorted with the first station. The primary reason for this relationship, as shown by Johnson and Meller (2002), is the blocking experienced by inductors as the number of inductors or induction stations increases. Moreover, space constraints may prohibit the installation of more induction stations. These constraints require that a sorter will only be able to sort a certain number of orders in a wave; this is denoted by $d_{\max}$ in Fig. 4.

Note that the relationship between the cost of the sorting system and the number of orders to be sorted in a wave is not linear; it is non-linear, in fact, strictly convex (for $I > 0$). The non-linearity is because of the diminishing returns in the induction rates due to increased levels of induction blocking in larger sorting systems. Therefore, the cost to increase the number of orders sorted increases rapidly as $d_{\max}$ is approached.

A detailed example of utilizing this sorter cost model is provided in the next section where we revisit the batch versus zone problem cost model presented in Section 4.

9. Cost model revisited through an example

We now present an illustrative example to indicate how our models can be used to estimate the system cost of batch picking and zone picking systems. We use the identical picking area and system parameters mentioned in Section 7 and summarize these values (along with other example parameter values) in Table 5. Note that the pack-rate of packers is typically higher in batch picking compared to zone picking as in the latter the packers may (i) have to sort items at the end of the sort-lane into individual customer orders if more than one order is assigned to each sort-lane, and (ii) have to walk from one sort-lane to another to perform the packing

Table 5
Notation used in the illustrative example

Parameters	Symbol	Value
Number of waves	w	12 per shift
Wavelength	l_w	40 min
Avg. # of items in an order	I_{avg}	
Small-sized orders		3 items
Large-sized orders		20 items
Total number of (narrow) aisles		16
Total number of pick-columns	n	800 (i.e., 50 pick-columns/aisle)
Pick-rate (batch picking)	r_p^b	150 items/h
Pick-rate (zone picking)	r_p^z	
2 pickers (zone-size = 8 aisles)		375 items/h
4 pickers (zone-size = 4 aisles)		469 items/h
8 pickers (zone-size = 2 aisles)		564 items/h
Speed of conveyor	s	100 items/min
Nominal induction rate	λ	20 items/min
# of orders per sort-lane	d_l	
Small-sized orders		5 orders/sort-lane
Large-sized orders		1 order/sort-lane
Pack-rate (batch picking)	r_k^b	180 orders/h
Pack-rate (zone picking)	r_k^z	100 orders/h
Imbalance cost	c_u	\$75 per item/year
Years of service		5 years
Interest rate		15%
Annualized cost of equipment	c_e	\$20,000 (batch) and \$40,000 (zone)
Annualized fixed cost of a sorter	c_f	\$96,405
Annualized cost of an induction station	c_s	\$17,798
Annualized cost of a sort-lane	c_l	\$371
Annual cost of a picker, inductor, or packer	$c_p, c_i, \text{ or } c_k$	\$40,000

operation. The cost components related to the sorter are representative values for a tilt-tray sorter. All cost values have been annualized (e.g., if the initial cost of an induction station is \$120,000, then its annualized cost for a 5 year service and at 15% interest rate is \$17,798).

For the system parameters mentioned in Table 5 we compare a sort-while-pick batch picking (SWPBP) to a simultaneous zone picking (SZP) system based on their system cost. In our comparison we consider three picking system throughput values (A_{sys}), low, medium, and high, and two order sizes, small and large. Based on the system throughput and order sizes, the five cost components (pickers, equipment, imbalance, sorting, and packers) are then estimated. Note that in a SWPBP system pickers sort the items into customer orders while they are picking; i.e., pickers carry along a compartmentalized pick-cart with enough compartments for the customer orders to be assigned to them. Table 6 summarizes the results of the SWPBP system.

For the SZP system we need to explicitly incorporate the sorter cost model. Fig. 5 shows the actual sorting system cost function, generated using the expressions developed in Section 8, for sorting 1000 small-sized orders in a 40-min wave in a SZP. The tuple (a,b) in Fig. 5 indicates the induction station number (#a) and the number of inductors (b) at that induction station, respectively. Therefore, (1,2) indicates that there is only one induction station, induction station #1, and it has two inductors assigned to it. Similarly, (1,3) and (2,2) indicates that there are two induction stations; induction station #1 has three inductors assigned to it and induction station #2 has two inductors assigned to it. This actual sorting system cost function is incorporated into the cost model for a SZP system. See Table 6 for a summary of the cost calculations.

For our illustrative example, we observe that a SWPBP system saves \$116,079 annually over a SZP system when the required order picking system throughput (A_{sys}) is low and order sizes are large. Though a majority of these savings comes from the SWPBP system not requiring a sorting system, around 10% of these savings are derived from the imbalance cost. However, as the system throughput increases, the savings derived by not requiring a sorting system in the SWPBP system is offset by the increase in the number of pickers required (from 5 to 13). As a result, for medium system throughput, in terms of system cost, the SWPBP and SZP

Table 6

Comparison of sort-while-pick batch picking (SWPBP) and simultaneous zone picking (SZP) systems for a given system throughput, order sizes, 12 waves/shift, and $I_{\text{avg}} = 3$ and 20 for small-sized and large-sized orders, respectively

	Low A_{sys}		Medium A_{sys}		High A_{sys}	
	SWPBP	SZP	SWPBP	SZP	SWPBP	SZP
Avg. daily orders, D						
Small-sized orders	2000	2000	5000	5000	12,000	12,000
Large-sized orders	300	300	750	750	1800	1800
Orders/wave, $D^w (= \lceil D/w \rceil)$						
small-sized orders	167	167	417	417	1000	1000
Large-sized orders	25	25	63	63	150	150
Items/wave, $I (= D^w I_{\text{avg}})$	500	500	1250	1250	3000	3000
Pick-rate/wave, $r_p^w (= (r_p I_w)/60)$	100	250	100	313	100	376
# of pickers, $P (= \lceil I/r_p \rceil)$	5	2	13	4	30	8
Picker blocking, $b(P)$	0%	–	2.2%	–	7.4%	–
(1) Cost of pickers, $\left(= \left[\frac{P}{1-b(P)/100} \right] c_p \right)$	\$200,000	\$80,000	\$560,000	\$160,000	\$1,320,000	\$320,000
(2) Equipment cost, c_e	\$20,000	\$40,000	\$20,000	\$40,000	\$20,000	\$40,000
Daily avg. # of items unfulfilled, U						
Small-sized orders	0	131	0	379	0	814
Large-sized orders	15	181	156	617	767	1371
(3) Imbalance cost, $U c_u$						
Small-sized orders	\$0	\$9825	\$0	\$28,425	\$0	\$61,050
Large-sized orders	\$975	\$13,575	\$11,700	\$46,275	\$0	\$102,825
# of sort-lanes, $n_l (= \lceil D^w/d_l \rceil)$						
Small-sized orders ($d_l = 5$)	–	34	–	84	–	200
Large-sized orders ($d_l = 1$)	–	25	–	63	–	150
# of induction stations, n_s	–	1	–	1	–	2
# of inductors, n_i	–	1	–	2	–	5
(4) Sorting cost, $c_s(I) (= c_r + n_l c_l + n_s c_s + n_i c_i)$						
Small-sized orders	–	\$166,818	–	\$225,368	–	\$406,202
Large-sized orders	–	\$163,479	–	\$217,577	–	\$387,652
Pack-rate/wave, $r_k^w (= (r_k I_w)/60)$ (orders/wave)	120	67	120	67	120	67
# of packers, $K (= \lceil D^w/r_k^w \rceil)$	2	3	4	7	9	15
(5) Cost of packers, $K c_k$	\$80,000	\$120,000	\$160,000	\$280,000	\$360,000	\$600,000
System cost, $C (= (1) + (2) + (3) + (4) + (5))$						
Small-sized orders	\$300,000	\$416,643	\$740,000	\$733,793	\$1,700,000	\$1,427,252
Large-sized orders	\$300,975	\$417,054	\$751,700	\$743,852	\$1,757,525	\$1,450,477

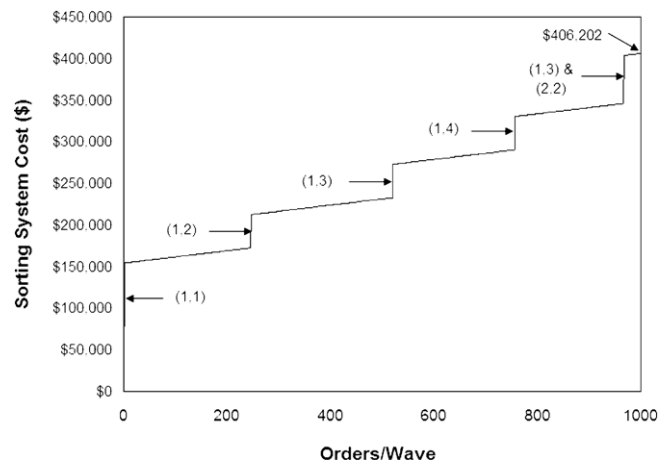


Fig. 5. An actual sorting system cost function for a sorter required to sort 1000 small-sized orders in a 40-min wave in a simultaneous zone picking system.

systems are essentially equally expensive. For high system throughput, the SWPBP system requires 30 pickers in the picking area, while the SZP system requires only 8 pickers. This savings in labor cost offsets the increase in the imbalance cost and the sorting system cost. In fact, the SZP system is now \$272,748 less expensive annually than the SWPBP system for large-sized orders.

In general, we see a progression of the most cost effective picking strategy; the SWPBP system is cost effective for low values of A_{sys} , both the SWPBP and SZP systems are equally cost effective for medium values of A_{sys} , while the SZP system is cost effective for high values of A_{sys} . This progression is dependent on the number of waves employed during the shift. As the number of waves decreases (i.e., the wavelength increases) the size of the sorting system, and its cost, will potentially increase. This means that as the number of waves decreases the SZP system may turn out to be more expensive even for medium values of A_{sys} .

Note that for the illustrative example we considered only one type of batch picking strategy along with one type of zone picking strategy (i.e., SWPBP and SZP). Our models are equally applicable for the other types of batch and zone picking strategies (i.e., pick-and-sort batch and progressive zone) with the appropriate changes in model parameters (e.g., pick-rate, pack-rate, and requirement of a sorter). For example, if we were to compare the SWPBP strategy with the pick-and-pass zone picking strategy, then we notice that neither of these strategies require a sorter. Consequently, the imbalance cost (apart from the labor cost) is expected to play a major role in differentiating the costs of the two systems, and hence, identifying the least expensive picking strategy.

10. Conclusions

Distribution centers (DCs) play a critical role in satisfying customer demand in global supply chains. The design of an order picking system in a DC is critical to both cost and service aspects of this relationship since selecting an appropriate picking strategy largely governs the cost of the picking system for a given throughput requirement. There are potentially four order picking strategies: discrete, batch, zone, and bucket brigade picking, which a designer can choose from. Our focus was on batch and zone picking strategies as they are frequently employed. We referred to this selection problem as the batch versus zone problem. Various factors, such as pick-rate, picker blocking, workload-imbalance, and sorting system requirement, that affect the batch versus zone problem were considered.

In light of the problem addressed, the specific contributions of this research are as follows:

- We developed a cost model to estimate the cost of a batch picking and a zone picking system. The cost model is comprised of the cost of pickers, equipment, imbalance, sorting system, and packers.
- We presented approaches to estimate the cost of pickers, equipment, and packers.
- To estimate the imbalance cost, we developed mathematical models, analogous to the dual bin-packing problem, considering a given assignment of items to the aisles. These models estimate the number of items unfulfilled, which can be used to estimate imbalance cost.
- We also developed a model to estimate cost of sorting system, in which we explicitly considered the cost of sort-lanes, unlike the approach presented in [Russell and Meller \(2003\)](#).

Based on our experiments, we found that workload-imbalance is greater in zone picking systems as compared to batch picking systems. Moreover, our experimental study showed that workload-imbalance is more prominent when (i) the order sizes increase, (ii) the item distribution is more non-uniform, and (iii) the number of waves increases.

In our sorting system cost model we indicated through a parametric cost function that the relationship between the sorting system cost and the number of orders is strictly convex. Finally, we demonstrated the use of our models through an illustrative example in which we compared sort-while-pick batch picking (SWPBP) and simultaneous zone picking (SZP) systems. For this example, a gradual progression of the most cost effective picking strategy based system throughput was observed: SWPBP for low system throughput, SWPBP or SZP for medium system throughput, and SZP for large system throughput.

An avenue for future research in this problem area would be to conduct a comprehensive experimental study focusing on the workload-imbalance issue by considering a larger set of parameter values as well as

considering multiple sets of customers orders. Moreover, considering the item-to-aisle assignment problem in zone picking in conjunction with the order batching problem may provide a broader perspective on the workload-imbalance issue in picking systems. It would also be helpful to develop a similar cost model for other order picking strategies (e.g., bucket brigade picking). Yet another area of future research would be to develop analytical models to select an appropriate picking system (manual, semi-automated, or automated).

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Appendix A. Experimental study

In our experimental study we consider an example problem where the picking area consists of 16 aisles, where each aisle is 100 ft long. An 8-h shift with twelve 40-min waves is considered. Two picking systems are considered, sort-while-pick batch picking and simultaneous zone picking. The pick-tour in a batch picking system covers all 16 aisles; however, the pick-tour in a zone picking system will cover only the number of aisles corresponding to the zone-size (i.e., the number of aisles in a zone). (In this example the zone-size in a zone picking system may be 1, 2, 4, or 8 aisles.) Smaller zone-sizes in a zone picking system, compared to that in a batch picking system, decrease the travel-time of pickers, which increases their pick-rates. Moreover, the pick-time in a simultaneous zone picking system is relatively small as pickers are not required to sort items while picking, unlike in a sort-while-pick batching picking system. In this example we assume that the expected value of pick-rate of pickers in a sort-while-pick batch picking system is 150 items/h. The zone-sizes in a simultaneous zone picking system, and the corresponding expected value of pick-rates of pickers, are as follows: (a) for a zone-size equal to 8 aisles (2 pickers, each assigned 8 aisles), the expected pick-rate is 375 items/h; (b) for a zone-size equal to 4 aisles (4 pickers, each assigned 4 aisles), the expected pick-rate is 469 items/h; and (c) for a zone-size equal to 2 aisles (8 pickers, each assigned 2 aisles), the expected pick-rate is 564 items/h.

The following parameters (and their values) were used in estimating these hourly pick-rates: (i) the walk-speed is 100 fpm, (ii) the pick-time in a sort-while-pick batch picking system is 8 s, while it is 6 s in a simultaneous zone picking system, (iii) the cross-aisle width is 10 ft, (iv) the time to drop-off the picked items at the depot at the end of each tour in a sort-while-pick batch picking system is 20 s, while the time to drop-off picked items at the end of each aisle in a simultaneous zone picking system is 11 s, and (v) the cart capacity is 75 items. So, for example, if the pickers follow a traversal routing policy in a sort-while-pick batch picking system, then they will have to tour the picking area 2 times in an hour. Accordingly, the total travel-time of pickers in such a system will be 39.12 min, and the total drop-off time is 0.67 min. Consequently, the time available for picking is 20.21 min, which at 8 s per pick gives 150 items/h. However, in a simultaneous zone picking system, with a zone-size equal to 8 aisles, the total travel-time of pickers (touring the picking area twice in an hour) is 19.56 min and the total drop-off time is 2.93 min. Therefore, the total time available for picking is 37.51 min, which at 6 s per pick gives 375 items/h. All other pick-rates were estimated in a similar manner.

Note that the order batching problem for batch picking requires only the number of items in an order. In contrast, the order batching problem for zone picking requires both the number and distribution of the items that comprise the orders over the zones. For example, an order may consist of 20 items. There exist numerous possibilities by which these 20 items may be distributed over the 16 aisles (e.g., 2, 1, 1, 1, 1, 2, 1, 1, 1, 1, 2, 1, 1, 1, and 2, or 5, 4, 0, 0, 0, 3, 0, 3, 0, 3, 0, 2, 0, 0, 0, and 0, etc.). That is, the items may be *nearly* uniformly distributed across the aisles or they may be *skewed*. Notice that the distribution of items in an order will not affect the order batching problem for batch picking, however, it may affect the order batching problem for zone picking.

Table 7

The percentage of items in customer orders generated according to Distribution D2 to be fulfilled from the 16 picking aisles

Aisle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
% of items	8	8	7.5	7.5	7	7	6.5	6.5	6	6	5.5	5.5	5	5	4.5	4.5

Table 8

Results for batch picking: small-sized orders and distributions D1 and D2

Orders (items)		FCFS	FCFS-50	FCFS-200	FFD	OBP-BP		LB
						Time (s)		
2000 (6000)	Obj.	5940	6000	6000	6000	6000	1436.62	6000
	Gap	1.00%	0.00%	0.00%	0.00%	0.00%		
	Unful.	60	0	0	0	0		0
5000 (15,000)	Obj.	14,846	15,000	15,000	15,000	14,949	86,400	15,000
	Gap	1.03%	0.00%	0.00%	0.00%	0.36%		
	Unful.	154	0	0	0	51		0
12,000 (36,000)	Obj.	35,580	36,000	36,000	36,000	NA	–	NA
	Gap	1.17%	0.00%	0.00%	0.00%	–		
	Unful.	420	0	0	0	–		–

Note: “NA” means CPLEX was unable to solve the problem on a computer with a 3.2 GHz processor and 2GB RAM; the number of binary decision variables for this problem was 4.32 million.

Table 9

Results for zone picking: small-sized orders and Distribution D1

Orders (items)		FCFS	FCFS-50	FCFS-200	FFD	OBP-ZP		LB
						Time (s)		
2000 (6000)	Obj.	5793	5905	5960	5987	5987	46.51	5987
	Gap	3.24%	1.37%	0.45%	0.00%	0.00%		
	Unful.	207	95	40	13	13		13
5000 (15,000)	Obj.	14,284	14,519	14,854	14,905	14,905	95.16	14,905
	Gap	4.17%	2.59%	0.34%	0.00%	0.00%		
	Unful.	716	481	146	95	95		95
12,000 (36,000)	Obj.	33,796	34,489	35,322	35,878	35,878	1974.93	35,878
	Gap	5.80%	3.87%	1.55%	0.00%	0.00%		
	Unful.	2204	1511	678	122	122		122

Table 10

Results for zone picking: small-sized orders and Distribution D2

Orders (items)		FCFS	FCFS-50	FCFS-200	FFD	OBP-ZP		LB
						Time (s)		
2000 (6000)	Obj.	5470	5503	5562	5689	5751	10.05	5751
	Gap	4.69%	4.14%	3.16%	1.04%	0.00%		
	Unful.	530	497	438	311	249		249
5000 (15,000)	Obj.	13,083	13,170	13,323	14,154	14,338	35.24	14,338
	Gap	8.42%	7.84%	6.81%	1.23%	0.00%		
	Unful.	1917	1830	1677	846	662		662
12,000 (36,000)	Obj.	30,882	31,475	31,740	34,108	34,494	150.18	34,494
	Gap	10.07%	8.41%	7.68%	1.08%	0.00%		
	Unful.	5118	4525	4260	1892	1506		1506

Table 11
Results for batch picking: large-sized orders and distributions D1 and D2

Orders (items)		FCFS	FCFS-50	FCFS-200	FFD	OBP-BP		LB
							Time (s)	
300 (6000)	Obj.	5401	5778	5805	5837	5987	86,400	6000
	Gap	9.98%	3.70%	3.25%	2.72%	0.22%		
	Unful.	599	222	195	163	13		0
750 (15,000)	Obj.	13,521	14,428	14,507	14,717	14,844	86,400	15,000
	Gap	9.86%	3.81%	3.29%	1.89%	1.04%		
	Unful.	1479	572	493	283	156		0
1800 (36,000)	Obj.	32,237	34,588	34,822	35,231	35,233	86,400	36,000
	Gap	10.45%	3.92%	3.27%	2.14%	2.18%		
	Unful.	3763	1412	1178	769	767		0

Table 12
Results for zone picking: large-sized orders and Distribution D1

Orders (items)		FCFS	FCFS-50	FCFS-200	FFD	OBP-ZP		LB
						Time (s)		
300 (6000)	Obj.	5654	5795	5851	5867	5981	10.68	5981
	Gap	5.46%	3.11%	2.17%	1.90%	0.00%		
	Unful.	346	205	149	133	19		19
750 (15,000)	Obj.	13,980	14,109	14,172	14,380	14,883	86,400	14,886
	Gap	6.08%	5.21%	4.79%	3.39%	0.02%		
	Unful.	1020	891	828	620	117		114
1800 (36,000)	Obj.	33,432	34,223	34,452	34,997	35,709	86,400	35,748
	Gap	6.46%	4.25%	3.61%	2.09%	0.11%		
	Unful.	2568	1777	1548	1003	291		252

Table 13
Results for zone picking: large-sized orders and Distribution D2

Orders (Items)		FCFS	FCFS-50	FCFS-200	FFD	OBP-ZP		LB
						Time (s)		
300 (6000)	Obj.	5415	5509	5509	5521	5658	25.19	5658
	Gap	4.06%	2.49%	2.49%	2.29%	0.00%		
	Unful.	585	491	491	479	342		342
750 (15,000)	Obj.	13,025	13,193	13,195	13,206	13,944	117.47	13,944
	Gap	6.17%	5.04%	5.03%	4.95%	0.00%		
	Unful.	1975	1807	1805	1794	1056		1056
1800 (36,000)	Obj.	30,566	31,024	31,214	31,661	33,549	785.14	33,549
	Gap	8.31%	7.04%	6.51%	5.26%	0.00%		
	Unful.	5434	4976	4786	4339	2451		2451

With this setup, we designed a small experiment to illustrate the effects of order sizes, item distribution over zones, and number of waves on the order batching problem and identified the resulting number of items unfulfilled. To analyze the effect of order sizes, we consider two customer order sizes: small and large. Small-sized orders (typical of direct-to-consumer DCs) contain on average three items with a range of 1–5 items, while large-sized (typical of retail DCs) orders contain on average 20 items with a range of 10–30 items. For each order size, we consider three picking system throughputs: low, medium, and high. In our experiments, for small-sized orders, low, medium, and high system throughputs correspond to 300 (6000), 750 (15,000), and

1800 (36,000) orders (items) per day, respectively. For large-sized orders, these values correspond to 2000 (6000), 5000 (15,000), and 12,000 (36,000) orders (items) per day, respectively. To analyze the effect of item distribution, we consider two distributions, D1 (a *uniform* distribution) and D2 (a *right-skewed* distribution). We generate one set of orders for each of these possibilities; a total of 12 sets of orders.

To analyze the effect of the number of waves, we consider six possibilities: one 8-h wave, two 4-h waves, four 2-h waves, eight 1-h waves, twelve 40-min waves, and sixteen 30-min waves. The selection of an appropriate number of waves is critical for any order picking system as its selection has cost implications, most notably in terms of the size of the sorting system. That is, the size of the sorting system is much larger in terms of the number of sort-lanes (and thus, expensive) in a system with four 1-h waves due to the relatively large number of orders to be sorted compared to a system with sixteen 30-min waves.

A lower bound (LB) was obtained for each of the problems by solving a linear relaxation of the OBP-BP and OBP-ZP models. The experimental outcomes are included in various tables and figures in [Appendices B and C](#).

Appendix B. Experimental study results (order sizes and item distribution)

This appendix summarizes the effect of order sizes and item distribution in orders on workload-imbalance for the example problem presented in Section 7. Note that we generated orders according to distributions D1 and D2. In our experiments Distribution D1 is uniform, while Distribution D2 is skewed and is as illustrated

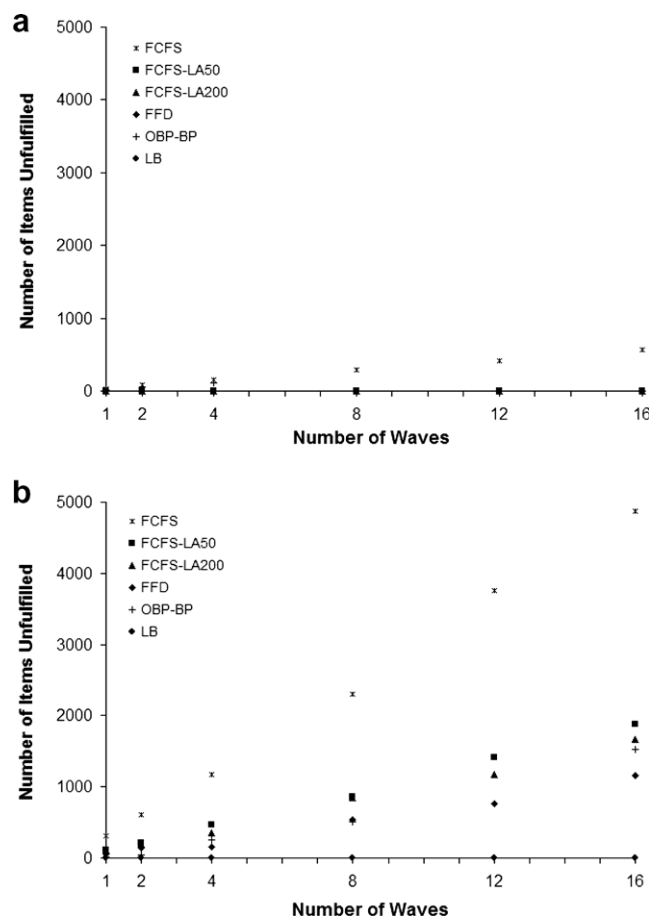


Fig. 6. Effect of wavelength on the number of items unfulfilled in a batch picking system: (a) small-sized orders and (b) large-sized orders.

in Table 7. Note that the number of items in each order generated according to distributions D1 and D2 is identical.

Because there are two order sizes (small and large), experiments were first conducted on small-sized orders (generated according to distributions D1 and D2) using the four order batching policies and the OBP-BP and OBP-ZP models presented earlier. These experiments were then repeated for large-sized orders. The order batching solutions obtained in terms of the total number of items included in the order batches by employing the four policies and the OBP-BP and OBP-ZP models are then compared to each other. The number of items unfulfilled is the difference of the total number of items to be fulfilled and the total number of items included in the batches.

Tables 8–10 summarize the results obtained for small-sized orders, while Tables 11–13 summarize the results for large-sized orders.

Appendix C. Experimental study results (waves)

This appendix summarizes the effect of the number of waves on workload-imbalance for the example problem presented in Section 7.

For our experimentation, both the batch and zone picking systems are considered. There are 12,000 small-sized orders and 1800 large-sized orders generated as per the Distribution D1 with a total of 36,000 items in each of them. Fig. 6 illustrates the number of items unfulfilled in small-sized and large-sized orders in a batch

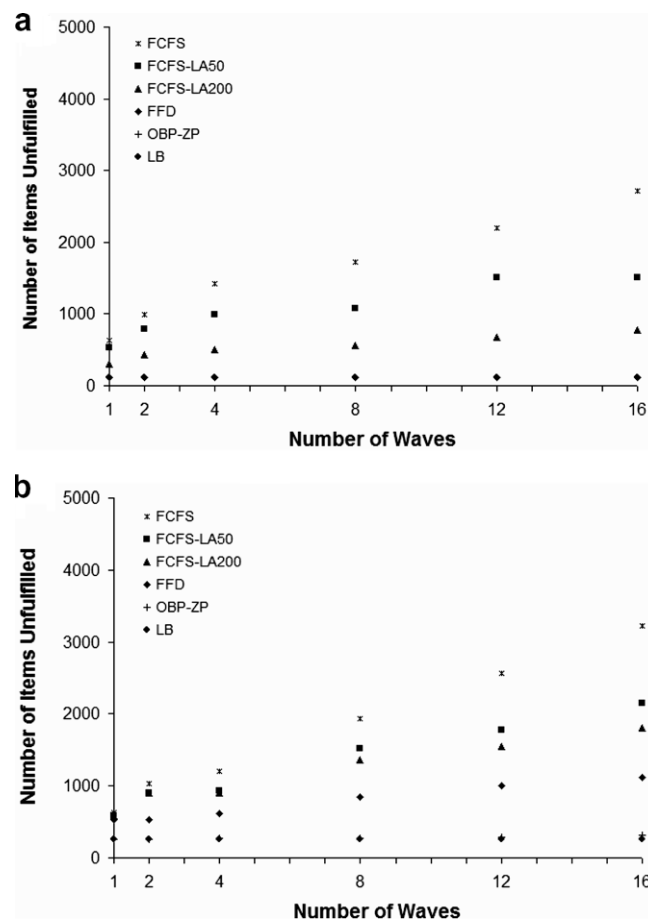


Fig. 7. Effect of wavelength on the number of items unfulfilled in a zone picking system: (a) small-sized orders and (b) large-sized orders.

picking system. Fig. 7 illustrates the number of items unfulfilled in small- and large-sized orders in a zone picking system.

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